Experiment 4

Numerical Investigation of Kármán Vortex Shedding

Aim

To capture alternating vortex formation downstream of a circular cylinder and estimate the corresponding Strouhal number from the unsteady force response.

Simulation Parameters

Parameter	Value
Coordinate system	Cylindrical
Radial grid points	80
Azimuthal grid points	160
Reynolds number	500
Iterations	1200
Lattice model	D2Q9 BGK Lattice Boltzmann

Numerical Method

The D2Q9 BGK lattice Boltzmann method is used with bounce-back treatment at the cylinder. Drag and lift are evaluated using momentum exchange, and the vortex-shedding frequency is estimated from the force history.

Boundary Conditions

A low-Mach inlet velocity and open outlet are used. The upper and lower boundaries use wall treatment, while no-slip at the cylinder is enforced with bounce-back.

Observations

Lift and drag histories, average drag, lift amplitude, estimated Strouhal number, density deviation, vorticity contours, and downstream wake periodicity were reviewed.

Results

Quantity	Approximate result
Maximum lift coefficient	+0.21
Minimum lift coefficient	–0.15
Lift peak-to-peak range	0.36
Initial maximum drag coefficient	20
Developed mean drag coefficient	1.5–1.6
Vortex-shedding period	7–8 (dimensionless time)
Estimated Strouhal number	0.13–0.14
Vorticity range	–9 to +7.5

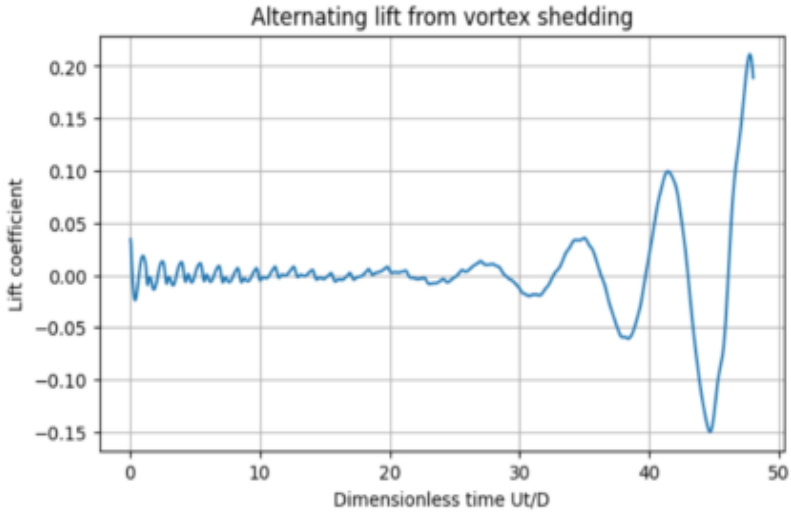
Conclusion

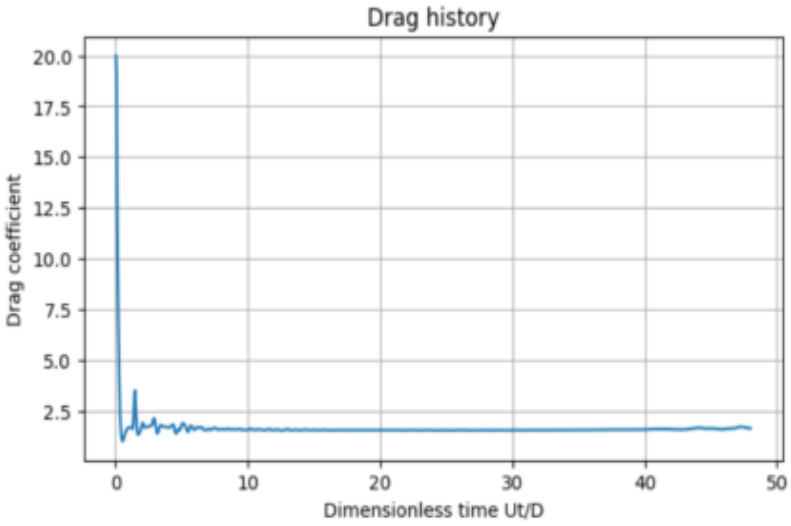
The lift signal develops alternating positive and negative peaks, consistent with vortices being shed from opposite sides of the cylinder. After a strong initial transient, the drag coefficient settles near 1.5–1.6. The vorticity field displays alternating structures downstream, forming a Kármán vortex street. The Strouhal estimate is approximate because the oscillations are still developing near the end of the simulated interval.

Original Simulation Figures

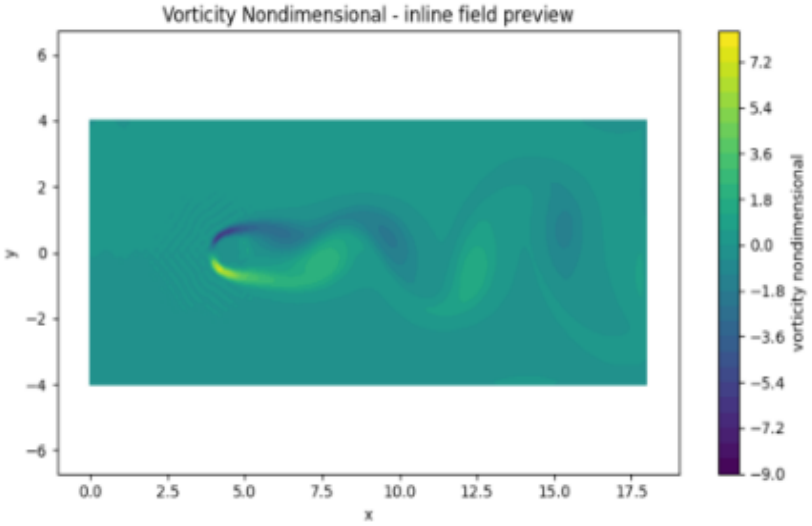
The following pages contain the plots and simulation outputs from the supplied report.

Original simulation output (source page 2)





Original simulation output (source page 3)



Results:

Parameter	**Result / Approx. Value**
Reynolds number	500
Grid	80 × 160
Total iterations	1200
Maximum positive lift coefficient,	≈ +0.21
Minimum lift coefficient,	≈ −0.15
Lift coefficient peak-to-peak	≈ 0.36
Initial maximum drag coefficient	≈ 20
Developed mean drag coefficient	≈ 1.5–1.6
Approx. vortex shedding period,	≈ 7–8
Estimated Strouhal number,	≈ 0.13–0.14
Vorticity range	≈ −9 to +7.5

Original simulation output (source page 4)

Parameter	Result / Approx. Value
Wake behaviour	Alternating periodic vortex shedding

Conclusion:

The lift coefficient initially exhibits small-amplitude oscillations around zero, indicating the development of the wake behind the cylinder. As the simulation progresses, the oscillations become increasingly pronounced. Towards the end of the simulation, the lift coefficient reaches a maximum of approximately +0.21 and a minimum of approximately −0.15, giving a peak-to-peak lift variation of approximately 0.36. This alternating lift behavior is associated with the periodic formation and shedding of vortices from opposite sides of the cylinder.

The drag history shows a strong initial transient. The drag coefficient reaches approximately 20 at the beginning of the simulation and then rapidly decreases as the flow field develops. After the initial transient, C_D settles to approximately 1.5–1.6, with relatively small fluctuations about this value. This indicates that the mean drag becomes nearly steady even though the wake continues to exhibit unsteady vortex shedding.

the corresponding estimated Strouhal number from the plotted oscillations is approximately 0.13–0.14. Since the oscillations are still developing toward the end of the simulation, this should be treated as an approximate graph-based value rather than an exact numerical result.

The vorticity contour provides further evidence of the periodic wake structure. The calculated non-dimensional vorticity varies approximately from −9 to +7.5, with the highest magnitudes concentrated near the cylinder surface and in the immediate wake. Alternating positive and negative vorticity regions can be observed downstream, forming a characteristic Kármán vortex street. The vortices become progressively more diffuse as they travel downstream because of mixing and viscous diffusion.

Overall, the results show that the simulation successfully captures the principal features of unsteady flow over a circular cylinder at Re=500: an initial force transient, stabilization of the mean drag near 1.5–1.6, alternating lift fluctuations with approximately 0.36 peak-to-peak amplitude, and periodic downstream vortex shedding. These observations are consistent with the report's stated focus on lift/drag histories, mean drag, lift amplitude, Strouhal number, vorticity, and wake periodicity.